

CSE30301: Basic Math for AI - Midterm Solution

Question 1 [30 points]

For each statement, the correct answer is as follows.

- 1.1 True.** Independence implies $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$, so $\text{Cov}(X, Y) = 0$.
- 1.2 False.** Zero covariance does not in general imply independence.
- 1.3 True.** $\text{Var}(\bar{X}) = \sigma^2/n$ for i.i.d. samples.
- 1.4 False.** A p-value is not $P(H_0 \text{ is true} \mid \text{data})$.
- 1.5 True.** An invertible matrix has rank full, so all singular values are positive.
- 1.6 True.** If A has full column rank, then $A^T A$ is invertible.
- 1.7 False.** This is true for symmetric matrices, but not for a general matrix.
- 1.8 False.** Least squares minimizes the residual norm, but does not always fit perfectly.
- 1.9 True.** In least squares, the residual is orthogonal to the column space of X .
- 1.10 True.** Increasing λ in ridge regression typically decreases variance.

Question 2 [20 points]

Let $X \sim \mathcal{N}(0, 1)$ and $Y = aX + X^2$.

2.1 Since $X \sim \mathcal{N}(0, 1)$,

$$\mathbb{E}[X] = 0, \quad \mathbb{E}[X^2] = 1, \quad \mathbb{E}[X^3] = 0.$$

Therefore,

$$\mathbb{E}[Y] = \mathbb{E}[aX + X^2] = a\mathbb{E}[X] + \mathbb{E}[X^2] = 0 + 1 = 1.$$

Also,

$$\mathbb{E}[XY] = \mathbb{E}[X(aX + X^2)] = a\mathbb{E}[X^2] + \mathbb{E}[X^3] = a \cdot 1 + 0 = a.$$

Hence,

$$\boxed{\mathbb{E}[X] = 0, \quad \mathbb{E}[Y] = 1, \quad \mathbb{E}[XY] = a.}$$

2.2 By definition,

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Using part 2.1,

$$\text{Cov}(X, Y) = a - (0)(1) = a.$$

Therefore,

$$\boxed{\text{Cov}(X, Y) = a.}$$

Question 3 [25 points]

Algorithm A: 11, 13, 15, 17, 19

Algorithm B: 14, 16, 18, 20, 22

3.1 The sample means are

$$\bar{X}_A = \frac{11 + 13 + 15 + 17 + 19}{5} = \frac{75}{5} = 15, \quad \bar{X}_B = \frac{14 + 16 + 18 + 20 + 22}{5} = \frac{90}{5} = 18.$$

So,

$$\boxed{\bar{X}_A = 15, \quad \bar{X}_B = 18.}$$

3.2 For Algorithm A, the deviations from the mean are $-4, -2, 0, 2, 4$, so

$$\sum_{i=1}^5 (x_i - \bar{X}_A)^2 = 16 + 4 + 0 + 4 + 16 = 40.$$

Thus,

$$s_A^2 = \frac{40}{5-1} = 10, \quad s_A = \sqrt{10}.$$

The same pattern holds for Algorithm B, so

$$s_B^2 = 10, \quad s_B = \sqrt{10}.$$

Therefore,

$$\boxed{s_A = \sqrt{10}, \quad s_B = \sqrt{10}.}$$

3.3 Using the approximate 95% confidence interval

$$\bar{X} \pm 2 \frac{s}{\sqrt{n}},$$

with $n = 5$ and $s = \sqrt{10}$,

$$\frac{s}{\sqrt{n}} = \frac{\sqrt{10}}{\sqrt{5}} = \sqrt{2} \approx 1.41.$$

So the margin is

$$2\sqrt{2} \approx 2(1.41) = 2.82.$$

Hence,

$$\text{A: } 15 \pm 2.82 \Rightarrow (12.18, 17.82),$$

$$\text{B: } 18 \pm 2.82 \Rightarrow (15.18, 20.82).$$

Therefore,

$$\boxed{\text{A: } 15 \pm 2.82, \quad \text{B: } 18 \pm 2.82.}$$

3.4 Algorithm B has a larger sample mean than Algorithm A. However, the two confidence intervals overlap, so based on these intervals alone, we should be cautious about claiming a clear statistically significant difference. Thus, B appears better on average, but the confidence intervals do not provide strong evidence of a clear difference.

Question 4 [30 points]

Let

$$A = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Since A is symmetric, $A^T A = A^2$.

4.1 First find the eigenvalues of A . The top-left 2×2 block

$$\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

has eigenvalues 4 and 2, and the third coordinate gives eigenvalue 1. Thus the eigenvalues of A are

$$4, 2, 1.$$

Therefore the eigenvalues of $A^T A = A^2$ are

$$4^2 = 16, \quad 2^2 = 4, \quad 1^2 = 1.$$

Hence,

Eigenvalues of $A^T A$: 16, 4, 1.

4.2 Since A is symmetric positive definite, its SVD coincides with its eigendecomposition. The orthonormal eigenvectors are:

$$v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad (\lambda = 4), \quad v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \quad (\lambda = 2), \quad v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad (\lambda = 1).$$

Therefore,

$$V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Because A is symmetric positive definite, we may take $U = V$. Hence one SVD is

$U = V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$

4.3 Using the SVD,

$$A^{-1} = V \Sigma^{-1} U^T, \quad \Sigma^{-1} = \begin{bmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Therefore,

$$A^{-1} = \begin{bmatrix} \frac{3}{8} & -\frac{1}{8} & 0 \\ -\frac{1}{8} & \frac{3}{8} & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Hence,

$$A^{-1} = \begin{bmatrix} \frac{3}{8} & -\frac{1}{8} & 0 \\ -\frac{1}{8} & \frac{3}{8} & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Question 5 [20 points]

Let

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

5.1 Let the columns of X be

$$x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad x_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}.$$

First,

$$\|x_1\| = \sqrt{3}, \quad q_1 = \frac{x_1}{\|x_1\|} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Next,

$$q_1^T x_2 = \frac{1}{\sqrt{3}}(0 + 1 + 2) = \sqrt{3}.$$

So

$$u_2 = x_2 - (q_1^T x_2)q_1 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

Since $\|u_2\| = \sqrt{2}$,

$$q_2 = \frac{u_2}{\|u_2\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

Therefore,

$$Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} \end{bmatrix}.$$

Also,

$$R = Q^T X = \begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{2} \end{bmatrix}.$$

Hence,

$$Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} \end{bmatrix}, \quad R = \begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{2} \end{bmatrix}.$$

5.2 We solve

$$\min_{\beta} \|X\beta - y\| = \min_{\beta} \|QR\beta - y\|.$$

Since Q has orthonormal columns, the least squares solution satisfies

$$R\hat{\beta} = Q^T y.$$

Compute

$$Q^T y = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{5}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}.$$

Thus we solve

$$\begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{2} \end{bmatrix} \hat{\beta} = \begin{bmatrix} \frac{5}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}.$$

From the second row,

$$\sqrt{2} \hat{\beta}_2 = \frac{1}{\sqrt{2}} \Rightarrow \hat{\beta}_2 = \frac{1}{2}.$$

From the first row,

$$\sqrt{3} \hat{\beta}_1 + \sqrt{3} \cdot \frac{1}{2} = \frac{5}{\sqrt{3}}.$$

Dividing by $\sqrt{3}$ gives

$$\hat{\beta}_1 + \frac{1}{2} = \frac{5}{3} \Rightarrow \hat{\beta}_1 = \frac{7}{6}.$$

Therefore,

$$\boxed{\hat{\beta} = \begin{bmatrix} \frac{7}{6} \\ \frac{1}{2} \end{bmatrix}}.$$

Question 6 [25 points]

Consider polynomial regression of degree d :

$$\hat{f}(x) = \sum_{k=0}^d \beta_k x^k.$$

6.1 Assume

$$y = f(x) + \epsilon, \quad \mathbb{E}[\epsilon] = 0, \quad \text{Var}(\epsilon) = \sigma^2.$$

Then

$$\mathbb{E}[(y - \hat{f}(x))^2] = \mathbb{E}[(f(x) + \epsilon - \hat{f}(x))^2].$$

Expanding,

$$= \mathbb{E}[(f(x) - \hat{f}(x))^2] + 2\mathbb{E}[(f(x) - \hat{f}(x))\epsilon] + \mathbb{E}[\epsilon^2].$$

Since $\mathbb{E}[\epsilon] = 0$ and $\mathbb{E}[\epsilon^2] = \sigma^2$,

$$\mathbb{E}[(y - \hat{f}(x))^2] = \mathbb{E}[(f(x) - \hat{f}(x))^2] + \sigma^2.$$

Now add and subtract $\mathbb{E}[\hat{f}(x)]$:

$$f(x) - \hat{f}(x) = (f(x) - \mathbb{E}[\hat{f}(x)]) + (\mathbb{E}[\hat{f}(x)] - \hat{f}(x)).$$

Therefore,

$$\mathbb{E}[(f(x) - \hat{f}(x))^2] = (f(x) - \mathbb{E}[\hat{f}(x)])^2 + \mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2],$$

because the cross term is zero. Hence,

$$\mathbb{E}[(y - \hat{f}(x))^2] = \underbrace{(\mathbb{E}[\hat{f}(x)] - f(x))^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2]}_{\text{Variance}} + \sigma^2.$$

So,

$$\boxed{\mathbb{E}[(y - \hat{f}(x))^2] = \text{Bias}^2 + \text{Variance} + \sigma^2.}$$

6.2 As d increases, the model becomes more flexible. In general,

$$\boxed{\text{bias decreases, while variance increases.}}$$

Small d tends to underfit, while large d can overfit.

6.3 Regularization penalizes overly large coefficients and reduces model complexity. In general,

$$\boxed{\text{regularization increases bias and decreases variance.}}$$

This can improve generalization performance.

Question 7 [15 points]

We start from

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Using the law of total expectation,

$$\mathbb{E}[X^2] = \mathbb{E}[\mathbb{E}[X^2 | Y]], \quad \mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X | Y]].$$

Also,

$$\text{Var}(X | Y) = \mathbb{E}[X^2 | Y] - (\mathbb{E}[X | Y])^2,$$

so

$$\mathbb{E}[X^2 | Y] = \text{Var}(X | Y) + (\mathbb{E}[X | Y])^2.$$

Taking expectation on both sides,

$$\mathbb{E}[X^2] = \mathbb{E}[\text{Var}(X | Y)] + \mathbb{E}[(\mathbb{E}[X | Y])^2].$$

Therefore,

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\ &= \mathbb{E}[\text{Var}(X | Y)] + \mathbb{E}[(\mathbb{E}[X | Y])^2] - (\mathbb{E}[\mathbb{E}[X | Y]])^2 \\ &= \mathbb{E}[\text{Var}(X | Y)] + \text{Var}(\mathbb{E}[X | Y]). \end{aligned}$$

Hence,

$$\boxed{\text{Var}(X) = \mathbb{E}[\text{Var}(X | Y)] + \text{Var}(\mathbb{E}[X | Y]).}$$

Question 8 [15 points]

Assume $P(Y = 0) = P(Y = 1) = \frac{1}{2}$,

$$\mathbb{E}[X | Y = 0] = 90, \quad \text{Var}(X | Y = 0) = 4,$$

$$\mathbb{E}[X | Y = 1] = 70, \quad \text{Var}(X | Y = 1) = 9.$$

8.1

$$\mathbb{E}[\text{Var}(X | Y)] = \frac{1}{2} \cdot 4 + \frac{1}{2} \cdot 9 = \frac{13}{2} = 6.5.$$

Hence,

$$\boxed{\mathbb{E}[\text{Var}(X | Y)] = 6.5.}$$

8.2 The random variable $\mathbb{E}[X | Y]$ takes values 90 and 70 with probability 1/2 each. Its mean is

$$\mathbb{E}[\mathbb{E}[X | Y]] = \frac{1}{2}(90) + \frac{1}{2}(70) = 80.$$

Therefore,

$$\text{Var}(\mathbb{E}[X | Y]) = \frac{1}{2}(90 - 80)^2 + \frac{1}{2}(70 - 80)^2 = \frac{1}{2}(100) + \frac{1}{2}(100) = 100.$$

Hence,

$$\boxed{\text{Var}(\mathbb{E}[X | Y]) = 100.}$$

8.3 By the law of total variance,

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid Y)] + \text{Var}(\mathbb{E}[X \mid Y]) = 6.5 + 100 = 106.5.$$

Hence,

$$\boxed{\text{Var}(X) = 106.5.}$$

Question 9 [20 points]

Let

$$X = wX_1 + (1 - w)X_2,$$

where X_1 and X_2 are independent, unbiased estimators of θ .

9.1 Using independence,

$$\text{Var}(X) = \text{Var}(wX_1 + (1 - w)X_2) = w^2\text{Var}(X_1) + (1 - w)^2\text{Var}(X_2).$$

Thus,

$$\boxed{\text{Var}(X) = w^2\sigma_1^2 + (1 - w)^2\sigma_2^2.}$$

9.2 Let

$$f(w) = w^2\sigma_1^2 + (1 - w)^2\sigma_2^2.$$

Differentiate:

$$f'(w) = 2w\sigma_1^2 - 2(1 - w)\sigma_2^2.$$

Setting $f'(w) = 0$ gives

$$2w\sigma_1^2 = 2(1 - w)\sigma_2^2.$$

Hence,

$$w\sigma_1^2 = \sigma_2^2 - w\sigma_2^2,$$

so

$$w(\sigma_1^2 + \sigma_2^2) = \sigma_2^2.$$

Therefore,

$$\boxed{w^* = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}.}$$

9.3 Since the weight on X_1 is larger when σ_1^2 is smaller, the estimator with smaller variance should receive a larger weight. Equivalently, the optimal weighting is inverse-variance weighting.